NAJILUL BAROKAH

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RESEARCH INTEREST

AI-driven researcher specializing in deep learning and quantum-inspired models for physiological signal analysis (EEG). Passionate about integrating AI with wearable devices for real-time mental health monitoring and cognitive state analysis in educational environments. Interested in leveraging multimodal data fusion to enhance personalized learning experiences and monitor student engagement/emotional well-being.

EDUCATION

Huazhong University of Science and Technology, China GPA: 88.33/100
M.Eng in Artificial Intelligence *Sep 2023 – Jun 2025*

Sunan Kalijaga State Islamic University, Indonesia GPA: 3.37/4.0
B.Ed in Physics Education *Sep 2017 – Dec 2021*

PUBLICATIONS

N. Barokah, Z. Fu, and J. Huang, "Advancing Emotion Recognition with Hybrid-Classical-Quantum Computing: A Deep Adaptation Network Approach," in 2025 44th Chinese Control Conference (CCC), Chongqing, China, 2025, pp. 9223-9228. doi: 10.23919/CCC64809.2025.11179167.

RELEVANT COURSEWORK & PROJECTS

Prince of Songkhla University, Thailand (Online)

Computational Physics *Grade A | Nov 2020 – Apr 2021*
- Developed Python scripts to model physics phenomena (e.g., wave equations, particle motion).

Seismology *Grade A | Jun – Oct 2021*
- Analyzed seismic signals to quantify earthquake magnitude and locate epicenters.

Geohazard *Grade B+ | Dec 2021 – Apr 2022*
- Studied disaster mitigation strategies for tsunamis, landslides, and volcanic eruptions.

RESEARCH EXPERIENCE

Hybrid Quantum-Classical Emotion Recognition *2023–2025*
- Developed a novel deep adaptation network with quantum circuits for emotion classification from physiological signals (EEG). Utilized Python, IBM Qiskit, and PyTorch to design variational quantum circuits that enhance feature extraction from complex physiological signals. Then , published the findings in the 2025 Chinese Control Conference (CCC), demonstrating the potential of quantum-inspired AI in biomedical applications.

Lab Assistant in Physics Practicum *2019–2021*
- Mentored 50+ students in fundamental physics and lab techniques.

PROFESSIONAL DEVELOPMENT

Workshop & Training on Gyrotron for Plasma Heating | Thailand Institute of Nuclear Technology | *September 2025*

- Studied high-frequency electromagnetic systems (gyrotron) and their applications in plasma heating, relevant to RF system design and EM wave propagation.
- Reviewed case studies from WEST (France) and EAST (China), emphasizing system integration and control.

Southeast Asian Workshop on Nuclear and Hadron Physics | Gadjah Mada University | *August 2025*
Actively participated in an international workshop featuring cutting-edge research presentations in nuclear and hadron physics from leading scientists worldwide. Gained valuable exposure to diverse experimental and theoretical approaches, expanding my understanding of the field's current global landscape and future directions.

TEACHING & LEADERSHIP

Vice Principal for Curriculum | SMK Praja Utama Karangpucung *June 2022–July 2023*
- Developed structured educational programs, emphasizing systematic design and implementation.

Physics Tutor | Avicenna Institute, Brilliant Insight Gandrungmangu *2019–2023*
- Trained students for national exams (UTBK) and physics subject, strengthening foundational skills in measurement and instrumentation.

TECHNICAL SKILLS

- Programming: Python (NumPy, TensorFlow, PyTorch, Qiskit)
- Quantum Computing: Quantum circuit design, variational quantum algorithms, IBM Qiskit
- Data Processing: Signal Pre-processing
- AI/ML: Deep learning, emotion recognition models, DNNs, transfer learning
- Tools: LaTeX, Sparkol Video Scribe (for science communication)

ADDITIONAL

- Languages: English (Professional), Indonesian (Native), Mandarin (Basic).
- Interests: Science outreach, quantum ethics, interdisciplinary AI applications.

REFERENCE

Prof. Huang Jian, Professor at School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. Email: huang_jan@mail.hust.edu.cn

Dr. Norma Sidik Risdianto, Ph.D, Assistant Professor at Faculty of Teaching and Education Science, UIN Sunan Kalijaga Yogyakarta. Email: norma.risdianto@uin-suka.ac.id

Asst. Prof. Dr. Boonyarit Chattong, Associate Dean for Research, Innovation and Public Relations, Prince of Songkla University. Email: boonyarit.ch@psu.ac.th